

# MATH 535: LINEAR ALGEBRA

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## 1. AUGUST 24, 2026

### 1.1. Fields.

**Definition 1** (Binary operation). *A binary operation on a set  $S$  is a function*

$$F: S \times S \longrightarrow S.$$

*Thus, for every  $x, y \in S$ , the value  $F(x, y)$  is again an element of  $S$ . We usually write  $x \cdot y$  or  $x + y$  in place of  $F(x, y)$ .*

**Example 1.** Addition on  $\mathbb{Z}$  is a binary operation: the map

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad (m, n) \longmapsto m + n$$

takes every pair of integers to an integer.

**Definition 2.** Let  $F$  be a binary operation on  $S$ . A subset  $T \subseteq S$  is closed under  $F$  if

$$F(x, y) \in T$$

whenever  $x, y \in T$ .

**Example 2.** The positive integers  $\mathbb{Z}_{>0}$  are closed under addition. The odd integers are not closed under addition, but they are closed under multiplication.

**Definition 3** (Field). Let  $\mathbb{F}$  be a set equipped with two binary operations, called addition and multiplication. The sum and product of  $a, b \in \mathbb{F}$  are denoted by  $a + b$  and  $ab$ , respectively. Suppose that for every  $a, b, c \in \mathbb{F}$ ,

- (i)  $a + b = b + a$ ;
- (ii)  $(a + b) + c = a + (b + c)$ ;
- (iii)  $ab = ba$ ;
- (iv)  $a(bc) = (ab)c$ ;
- (v)  $a(b + c) = ab + ac$ .

Suppose further that

- (vi)  $\mathbb{F}$  contains an additive identity  $0$  and a multiplicative identity  $1$ , so that

$$a + 0 = a \quad \text{and} \quad 1a = a$$

for every  $a \in \mathbb{F}$ ;

- (vii)  $1 \neq 0$ ;
- (viii) every  $a \in \mathbb{F}$  has an additive inverse  $-a \in \mathbb{F}$  such that  $a + (-a) = 0$ ;
- (ix) every nonzero  $a \in \mathbb{F}$  has a multiplicative inverse  $a^{-1} \in \mathbb{F}$  such that  $aa^{-1} = 1$ .

Then  $\mathbb{F}$  is called a field.

**Example 3.** *The integers  $\mathbb{Z}$  do not form a field: for example, 2 has no multiplicative inverse in  $\mathbb{Z}$ . The sets  $\mathbb{Q}$ ,  $\mathbb{R}$ , and  $\mathbb{C}$ , with their usual operations, are fields.*

There are also finite fields. For example,

$$\mathbb{F}_2 = \{0, 1\} \quad \text{and} \quad \mathbb{F}_3 = \{0, 1, 2\},$$

where the elements are the possible remainders upon division by 2 and 3, respectively. Addition and multiplication in  $\mathbb{F}_3$  are given by the following tables:

$$\begin{array}{c|ccc} + & 0 & 1 & 2 \\ \hline 0 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 \end{array} \quad \begin{array}{c|ccc} \cdot & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 \\ 2 & 0 & 2 & 1 \end{array}.$$

More generally, if  $p$  is prime, then

$$\mathbb{F}_p = \{0, 1, \dots, p-1\},$$

with addition and multiplication performed modulo  $p$ , is a field.

**Lemma 1.** *If  $a$  is an element of a field  $\mathbb{F}$ , then  $a \cdot 0 = 0$ .*

*Proof.* Since  $0 + 0 = 0$ , distributivity gives

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding the additive inverse of  $a \cdot 0$  to both sides yields  $0 = a \cdot 0$ .  $\square$

## 2. AUGUST 26, 2026

**2.1. Further properties of fields.** The preceding lecture showed that  $a \cdot 0 = 0$  for every element  $a$  of a field. We now obtain the zero-product property.

**Proposition 1** (Zero-product property). *Let  $a, b \in \mathbb{F}$ . If  $ab = 0$ , then either  $a = 0$  or  $b = 0$ .*

*Proof.* Suppose that  $ab = 0$  and  $a \neq 0$ . Since  $\mathbb{F}$  is a field,  $a^{-1}$  exists. Therefore

$$0 = a^{-1}0 = a^{-1}(ab) = (a^{-1}a)b = 1b = b.$$

Thus  $a = 0$  or  $b = 0$ . □

**2.2. Characteristic of a field.** For a positive integer  $n$  and  $a \in \mathbb{F}$ , write

$$na = \underbrace{a + a + \cdots + a}_{n \text{ terms}}.$$

**Definition 4** (Characteristic). *Let  $\mathbb{F}$  be a field. If  $n1 = 0$  for some positive integer  $n$ , then the least such  $n$  is called the characteristic of  $\mathbb{F}$ , denoted  $\text{char}(\mathbb{F})$ . If  $n1 \neq 0$  for every positive integer  $n$ , then  $\mathbb{F}$  has characteristic zero.*

The fields  $\mathbb{Q}$ ,  $\mathbb{R}$ , and  $\mathbb{C}$  have characteristic zero. For every prime  $p$ , the field  $\mathbb{F}_p$  has characteristic  $p$ . The rational function field  $\mathbb{F}_p(t)$  is an infinite field of characteristic  $p$ .

**Proposition 2.** *If  $\mathbb{F}$  has positive characteristic  $r$ , then  $r$  is prime.*

*Proof.* We have  $r \neq 1$ , since  $1 \neq 0$ . Suppose that  $r$  is composite, so

$$r = st$$

for integers  $s, t$  satisfying  $1 < s, t < r$ . Distributivity gives

$$(s1)(t1) = (st)1 = r1 = 0.$$

By the zero-product property, either  $s1 = 0$  or  $t1 = 0$ . Both alternatives contradict the minimality of  $r$ . Hence  $r$  is prime. □

**Proposition 3.** *If  $\text{char}(\mathbb{F}) = p > 0$ , then*

$$pa = 0$$

*for every  $a \in \mathbb{F}$ .*

*Proof.* Since  $p1 = 0$ , associativity and distributivity imply

$$pa = p(1a) = (p1)a = 0a = 0.$$

□

**Proposition 4.** *Every finite field has positive characteristic.*

*Proof.* Let  $\mathbb{F}$  be finite. The sequence

$$1, \quad 2 \cdot 1, \quad 3 \cdot 1, \quad \dots$$

cannot consist of distinct elements. Hence there are positive integers  $m > n$  such that

$$m1 = n1.$$

Adding the additive inverse of  $n1$  to both sides gives

$$(m - n)1 = 0.$$

Thus  $\mathbb{F}$  has positive characteristic. □

### 2.3. Algebraically closed fields.

**Definition 5** (Algebraically closed field). *A field  $\mathbb{F}$  is algebraically closed if every nonconstant polynomial  $f \in \mathbb{F}[x]$  has a root in  $\mathbb{F}$ .*

The field  $\mathbb{Q}$  is not algebraically closed, because  $x^2 - 2$  has no root in  $\mathbb{Q}$ . The field  $\mathbb{R}$  is not algebraically closed, because  $x^2 + 1$  has no real root. By the fundamental theorem of algebra,  $\mathbb{C}$  is algebraically closed.

**Proposition 5.** *No finite field is algebraically closed.*

*Proof.* Let

$$\mathbb{F} = \{a_1, \dots, a_q\}$$

be finite. Consider the nonconstant polynomial

$$f(x) = 1 + \prod_{i=1}^q (x - a_i).$$

For each  $i = 1, \dots, q$ ,

$$f(a_i) = 1 \neq 0.$$

Thus  $f$  has no root in  $\mathbb{F}$ , so  $\mathbb{F}$  is not algebraically closed. □

**2.4. Vectors and vector spaces.** The spaces  $\mathbb{R}^2$ ,  $\mathbb{R}^3$ , and, more generally,

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\}$$

are the motivating examples of vector spaces.

**Definition 6** (Vector space). *Let  $\mathbb{F}$  be a field. A vector space over  $\mathbb{F}$  is a set  $V$  equipped with two operations,*

$$V \times V \longrightarrow V, \quad (u, v) \longmapsto u + v,$$

and

$$\mathbb{F} \times V \longrightarrow V, \quad (\alpha, v) \longmapsto \alpha v,$$

satisfying the following axioms for all  $u, v, w \in V$  and  $\alpha, \beta \in \mathbb{F}$ :

$$(V1) \quad u + (v + w) = (u + v) + w;$$

$$(V2) \quad u + v = v + u;$$

$$(V3) \quad \text{there is an element } 0 \in V \text{ such that } v + 0 = 0 + v = v;$$

$$(V4) \quad \text{for every } v \in V, \text{ there is an element } -v \in V \text{ such that } v + (-v) = (-v) + v = 0;$$

$$(V5) \quad 1v = v;$$

$$(V6) \quad (\alpha\beta)v = \alpha(\beta v);$$

$$(V7) \quad \alpha(u + v) = \alpha u + \alpha v;$$

$$(V8) \quad (\alpha + \beta)v = \alpha v + \beta v.$$

The elements of  $V$  are called *vectors*, and the elements of  $\mathbb{F}$  are called *scalars*.

### 3. AUGUST 28, 2026

#### 3.1. Examples of vector spaces.

**Example 4** (Coordinate spaces). *Let  $\mathbb{F}$  be a field. The set*

$$\mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{F} \right\}$$

*is a vector space over  $\mathbb{F}$  under componentwise addition and scalar multiplication:*

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \alpha \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix}.$$

In particular,  $\mathbb{R}^n$  is a vector space over  $\mathbb{R}$  and  $\mathbb{C}^n$  is a vector space over  $\mathbb{C}$ .

**Example 5** (Matrix spaces). *The set  $M_{m \times n}(\mathbb{F})$  of  $m \times n$  matrices with entries in  $\mathbb{F}$  is a vector space over  $\mathbb{F}$ . Addition and scalar multiplication are defined entrywise:*

$$[a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}], \quad \alpha[a_{ij}] = [\alpha a_{ij}].$$

**Example 6** (Polynomial spaces). *Let*

$$\mathcal{P}_n(\mathbb{F}) = \{a_0 + a_1t + \cdots + a_nt^n : a_0, \dots, a_n \in \mathbb{F}\}.$$

*This is the vector space over  $\mathbb{F}$  of polynomials of degree at most  $n$ . Its operations are*

$$\begin{aligned} (a_0 + a_1t + \cdots + a_nt^n) + (b_0 + b_1t + \cdots + b_nt^n) \\ = (a_0 + b_0) + (a_1 + b_1)t + \cdots + (a_n + b_n)t^n, \end{aligned}$$

*and*

$$\alpha(a_0 + a_1t + \cdots + a_nt^n) = \alpha a_0 + \alpha a_1t + \cdots + \alpha a_nt^n.$$

### 3.2. Linear combinations and spanning sets.

**Definition 7** (Linear combination and span). *Let  $V$  be a vector space over  $\mathbb{F}$  and let  $v_1, \dots, v_p \in V$ . A linear combination of these vectors is a vector of the form*

$$\alpha_1 v_1 + \cdots + \alpha_p v_p, \quad \alpha_1, \dots, \alpha_p \in \mathbb{F}.$$

*The set of all their linear combinations is called their span:*

$$\text{span}\{v_1, \dots, v_p\} = \{\alpha_1 v_1 + \cdots + \alpha_p v_p : \alpha_1, \dots, \alpha_p \in \mathbb{F}\}.$$

*A linear combination is trivial if all of its coefficients are zero.*

In  $\mathbb{R}^2$ , let

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$\text{span}\{e_1, e_2\} = \mathbb{R}^2,$$

because

$$\begin{bmatrix} x \\ y \end{bmatrix} = xe_1 + ye_2.$$

If

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

then  $\{e_1, e_2, v\}$  also spans  $\mathbb{R}^2$ , since

$$\begin{bmatrix} x \\ y \end{bmatrix} = xe_1 + ye_2 + 0v.$$

The additional vector is unnecessary, which motivates the question of redundancy in a spanning set.

**Definition 8** (Spanning set). *A collection  $\{v_1, \dots, v_p\} \subseteq V$  is a spanning set for  $V$  if every  $v \in V$  can be written as a linear combination*

$$v = \alpha_1 v_1 + \dots + \alpha_p v_p.$$

*Equivalently,*

$$\text{span}\{v_1, \dots, v_p\} = V.$$

### 3.3. Linear independence.

**Definition 9** (Linear independence). *A collection  $\{v_1, \dots, v_p\} \subseteq V$  is linearly independent if*

$$\alpha_1 v_1 + \dots + \alpha_p v_p = 0$$

*implies*

$$\alpha_1 = \dots = \alpha_p = 0.$$

*In other words, only the trivial linear combination of  $v_1, \dots, v_p$  equals zero. A collection that is not linearly independent is called linearly dependent.*

**Example 7.** *Consider the vectors*

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

in  $\mathbb{R}^3$ . To test linear independence, solve

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

This gives the system

$$\begin{cases} \alpha + 2\beta + 3\gamma = 0, \\ \alpha + \gamma = 0, \\ \beta + \gamma = 0. \end{cases}$$

Taking  $\gamma = 1$  gives  $\alpha = \beta = -1$ , so

$$-v_1 - v_2 + v_3 = 0.$$

Thus the vectors are linearly dependent; equivalently,  $v_3 = v_1 + v_2$ .

The following result precisely characterizes redundancy in a finite collection of vectors.

**Proposition 6.** *Let  $v_1, \dots, v_p \in V$ . The collection  $\{v_1, \dots, v_p\}$  is linearly dependent if and only if one of its vectors is a linear combination of the others. Equivalently, there is an index  $k$  and scalars  $\beta_j \in \mathbb{F}$  such that*

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j.$$

*Proof.* Suppose first that  $\{v_1, \dots, v_p\}$  is linearly dependent. Then there are scalars  $\alpha_1, \dots, \alpha_p$ , not all zero, such that

$$\alpha_1 v_1 + \dots + \alpha_p v_p = 0.$$

Choose  $k$  such that  $\alpha_k \neq 0$ . Since  $\mathbb{F}$  is a field,  $\alpha_k^{-1}$  exists, and we can solve for  $v_k$ :

$$v_k = - \sum_{\substack{j=1 \\ j \neq k}}^p \alpha_k^{-1} \alpha_j v_j.$$

Thus  $v_k$  is a linear combination of the other vectors.

Conversely, suppose that for some  $k$ ,

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j.$$

Moving all terms to one side gives

$$v_k - \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j = 0.$$

This is a nontrivial linear combination of  $v_1, \dots, v_p$ , because the coefficient of  $v_k$  is  $1 \neq 0$ . Hence the collection is linearly dependent.  $\square$

#### 4. AUGUST 31, 2026

##### 4.1. Bases and coordinates.

**Definition 10** (Basis). *A collection of vectors  $\{v_1, \dots, v_n\}$  in a vector space  $V$  is a basis for  $V$  if it is linearly independent and spans  $V$ .*

The two conditions in the definition can be combined into a statement about representing vectors uniquely.

**Proposition 7.** *A collection  $\{v_1, \dots, v_n\}$  is a basis for  $V$  if and only if every vector  $v \in V$  has a unique representation of the form*

$$v = a_1 v_1 + \dots + a_n v_n, \quad a_1, \dots, a_n \in \mathbb{F}.$$

*When  $\{v_1, \dots, v_n\}$  is a basis, the scalars  $a_1, \dots, a_n$  are called the coordinates of  $v$  with respect to this basis.*

*Proof.* Suppose first that  $\{v_1, \dots, v_n\}$  is a basis. Since the collection spans  $V$ , every  $v \in V$  has a representation

$$v = a_1 v_1 + \dots + a_n v_n.$$

To prove uniqueness, suppose that

$$v = \tilde{a}_1 v_1 + \dots + \tilde{a}_n v_n$$

is another such representation. Subtracting the two equations gives

$$(a_1 - \tilde{a}_1)v_1 + \dots + (a_n - \tilde{a}_n)v_n = 0.$$

Because  $\{v_1, \dots, v_n\}$  is linearly independent,

$$a_i - \tilde{a}_i = 0$$

for each  $i$ . Thus  $a_i = \tilde{a}_i$  for every  $i$ , and the representation is unique.

Conversely, suppose every  $v \in V$  has a unique representation as a linear combination of  $v_1, \dots, v_n$ . The existence of such a representation for every  $v$  shows that the collection spans  $V$ . Now suppose

$$a_1v_1 + \dots + a_nv_n = 0.$$

The zero vector also has the representation

$$0 = 0v_1 + \dots + 0v_n.$$

Uniqueness therefore gives  $a_1 = \dots = a_n = 0$ . Hence the collection is linearly independent, so it is a basis for  $V$ .  $\square$

**Example 8** (The standard basis of  $\mathbb{F}^n$ ). *For  $i = 1, \dots, n$ , let  $e_i \in \mathbb{F}^n$  be the vector whose  $i$ th coordinate is 1 and whose other coordinates are 0. Thus*

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

*Every vector in  $\mathbb{F}^n$  has the unique representation*

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1e_1 + x_2e_2 + \dots + x_ne_n.$$

*Consequently,  $\{e_1, \dots, e_n\}$  is a basis for  $\mathbb{F}^n$ , called its standard basis.*

**Example 9** (The standard basis of  $\mathcal{P}_n(\mathbb{F})$ ). *Recall that  $\mathcal{P}_n(\mathbb{F})$  is the vector space of polynomials over  $\mathbb{F}$  of degree at most  $n$ . Each polynomial has a unique expression*

$$p(t) = a_0 + a_1t + \dots + a_nt^n.$$

Hence

$$\{1, t, t^2, \dots, t^n\}$$

is a basis for  $\mathcal{P}_n(\mathbb{F})$ , called its standard basis.

**Proposition 8.** *Every finite nonempty spanning set for a vector space contains a basis.*

*Proof.* Let  $\{v_1, \dots, v_p\}$  span  $V$ . If this collection is linearly independent, then it is already a basis. Otherwise, by Proposition 6, some  $v_k$  is a linear combination of the other vectors. It follows that removing  $v_k$  does not change the span:

$$\text{span}\{v_1, \dots, v_p\} = \text{span}\{v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_p\}.$$

If the remaining collection is linearly independent, it is a basis. If not, repeat the same argument. The original spanning set is finite, so this process must terminate. The final collection still spans  $V$  and is linearly independent, and is therefore a basis contained in the original spanning set.  $\square$

**4.2. Linear transformations.** A *transformation*, also called a map or function, from a set  $X$  to a set  $Y$  is a rule  $T: X \rightarrow Y$  that assigns to every  $x \in X$  an element  $T(x) \in Y$ .

**Definition 11** (Linear transformation). *Let  $V$  and  $W$  be vector spaces over the same field  $\mathbb{F}$ . A transformation  $T: V \rightarrow W$  is linear if*

- (i)  $T(u + v) = T(u) + T(v)$  for all  $u, v \in V$ ;
- (ii)  $T(\alpha u) = \alpha T(u)$  for all  $u \in V$  and  $\alpha \in \mathbb{F}$ .

Equivalently,  $T: V \rightarrow W$  is linear if

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all  $u, v \in V$  and  $\alpha, \beta \in \mathbb{F}$ .

**Example 10** (Reflection). *Define  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  by*

$$T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}.$$

Geometrically,  $T$  reflects each point across the first coordinate axis. To check additivity, observe that

$$\begin{aligned} T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) &= T\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 + y_1 \\ -x_2 - y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ -y_2 \end{bmatrix} \\ &= T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + T\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \end{aligned}$$

For  $\alpha \in \mathbb{R}$ ,

$$\begin{aligned} T\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) &= T\begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha x_1 \\ -\alpha x_2 \end{bmatrix} \\ &= \alpha \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} = \alpha T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \end{aligned}$$

Thus  $T$  is linear.

**Example 11** (The differentiation operator). Let  $n \geq 1$ ,  $V = \mathcal{P}_n(\mathbb{F})$ , and  $W = \mathcal{P}_{n-1}(\mathbb{F})$ . Define

$$T: V \longrightarrow W, \quad T(p) = p'.$$

Since

$$(p + q)' = p' + q' \quad \text{and} \quad (\alpha p)' = \alpha p',$$

the differentiation operator  $T$  is linear.

5. SEPTEMBER 2, 2026

**5.1. Linear transformations determined by a basis.** The following result shows that prescribing the values of a linear transformation on a basis completely determines the transformation.

**Lemma 2** (Linear map lemma). *Suppose that  $\{v_1, \dots, v_n\}$  is a basis for  $V$ , and let  $w_1, \dots, w_n$  be arbitrary vectors in  $W$ . Then there exists a unique linear transformation  $T: V \rightarrow W$  such that*

$$T(v_k) = w_k$$

for each  $k = 1, \dots, n$ .

*Proof.* Every  $v \in V$  has a unique representation

$$v = \sum_{i=1}^n c_i v_i.$$

Define

$$T(v) = \sum_{i=1}^n c_i w_i.$$

The uniqueness of the coordinates  $c_1, \dots, c_n$  shows that this definition is well-defined. In particular, taking  $v = v_k$  gives  $T(v_k) = w_k$ .

To prove that  $T$  is linear, write

$$u = \sum_{i=1}^n a_i v_i \quad \text{and} \quad v = \sum_{i=1}^n c_i v_i.$$

Then

$$\begin{aligned} T(u+v) &= T\left(\sum_{i=1}^n (a_i + c_i)v_i\right) \\ &= \sum_{i=1}^n (a_i + c_i)w_i \\ &= \sum_{i=1}^n a_i w_i + \sum_{i=1}^n c_i w_i \\ &= T(u) + T(v). \end{aligned}$$

Similarly, for every  $\alpha \in \mathbb{F}$ ,

$$\begin{aligned} T(\alpha v) &= T\left(\sum_{i=1}^n \alpha c_i v_i\right) \\ &= \sum_{i=1}^n \alpha c_i w_i = \alpha \sum_{i=1}^n c_i w_i = \alpha T(v). \end{aligned}$$

Thus  $T$  is linear.

It remains to prove uniqueness. Suppose that  $T': V \rightarrow W$  is another linear transformation satisfying  $T'(v_k) = w_k$  for every  $k$ . If  $v = \sum_{i=1}^n c_i v_i$ , then

$$T'(v) = \sum_{i=1}^n c_i T'(v_i) = \sum_{i=1}^n c_i w_i = T(v).$$

Hence  $T' = T$ . □

**5.2. Matrix representations of linear transformations.** An *ordered basis* is a basis whose vectors are listed in a fixed order. Let

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m)$$

be ordered bases for  $V$  and  $W$ , respectively. For  $v = x_1 v_1 + \dots + x_n v_n$ , write

$$[v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

for the coordinate vector of  $v$  with respect to  $\mathcal{B}$ .

**Theorem 1** (Matrix representation theorem). *Let  $V$  and  $W$  be vector spaces over  $\mathbb{F}$  with ordered bases  $\mathcal{B} = (v_1, \dots, v_n)$  and  $\mathcal{C} = (w_1, \dots, w_m)$ . For every linear transformation  $T: V \rightarrow W$ , there is a unique matrix*

$$A = [A_{ij}] = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \in M_{m \times n}(\mathbb{F})$$

such that

$$T(v_j) = \sum_{i=1}^m A_{ij} w_i$$

for each  $j = 1, \dots, n$ . Conversely, every matrix  $A \in M_{m \times n}(\mathbb{F})$  determines a unique linear transformation  $T: V \rightarrow W$  through this formula.

*Proof.* Let  $T: V \rightarrow W$  be linear. Because  $\mathcal{C}$  is a basis for  $W$ , each vector  $T(v_j)$  has a unique representation

$$T(v_j) = A_{1j}w_1 + \cdots + A_{mj}w_m = \sum_{i=1}^m A_{ij}w_i.$$

Use these coordinate vectors as the columns of  $A$ :

$$A = \left[ \begin{array}{c|ccc|c} & & & & \\ & | & & & | \\ [T(v_1)]_{\mathcal{C}} & & \cdots & & [T(v_n)]_{\mathcal{C}} \\ & | & & & | \end{array} \right].$$

This constructs the required matrix. If  $B = [B_{ij}]$  were another matrix with the same property, then for every  $j$ ,

$$\sum_{i=1}^m B_{ij}w_i = T(v_j) = \sum_{i=1}^m A_{ij}w_i.$$

The uniqueness of coordinates with respect to  $\mathcal{C}$  gives  $B_{ij} = A_{ij}$  for every  $i$  and  $j$ . Hence  $B = A$ .

Conversely, let  $A = [A_{ij}]$  be an arbitrary  $m \times n$  matrix and define

$$\tilde{w}_j = \sum_{i=1}^m A_{ij}w_i \quad (j = 1, \dots, n).$$

By the linear map lemma, there is a unique linear transformation  $T: V \rightarrow W$  such that  $T(v_j) = \tilde{w}_j$  for every  $j$ . Thus  $A$  determines a unique linear transformation with the required property.  $\square$

If

$$v = x_1v_1 + \cdots + x_nv_n,$$

then linearity gives

$$\begin{aligned} T(v) &= \sum_{j=1}^n x_j T(v_j) \\ &= \sum_{j=1}^n x_j \sum_{i=1}^m A_{ij}w_i \\ &= \sum_{i=1}^m \left( \sum_{j=1}^n A_{ij}x_j \right) w_i. \end{aligned}$$

Consequently,

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Thus multiplication by the matrix  $A$  computes the coordinates of the image of a vector from the coordinates of the original vector.

6. SEPTEMBER 4, 2026

**6.1. Matrix–vector multiplication.** Let  $V$  and  $W$  be vector spaces over  $\mathbb{F}$  with ordered bases

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m),$$

respectively, and let  $T: V \rightarrow W$  be linear. Recall that the matrix representing  $T$  with respect to these bases is

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix},$$

where

$$T(v_j) = A_{1j}w_1 + \cdots + A_{mj}w_m.$$

Thus the  $j$ th column of  $A$  is the coordinate vector  $[T(v_j)]_{\mathcal{C}}$ .

Suppose that

$$v = x_1v_1 + \cdots + x_nv_n, \quad [v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

By linearity,

$$\begin{aligned} [T(v)]_{\mathcal{C}} &= x_1[T(v_1)]_{\mathcal{C}} + \cdots + x_n[T(v_n)]_{\mathcal{C}} \\ &= x_1 \begin{bmatrix} A_{11} \\ A_{21} \\ \vdots \\ A_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} A_{1n} \\ A_{2n} \\ \vdots \\ A_{mn} \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} A_{11}x_1 + \cdots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n \end{bmatrix}.$$

This motivates the following definition.

**Definition 12** (Matrix–vector product). *Let  $A = [A_{ij}] \in M_{m \times n}(\mathbb{F})$ , and write its columns as  $A_1, \dots, A_n$ . For*

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{F}^n,$$

*define*

$$Ax = x_1A_1 + \cdots + x_nA_n.$$

*Equivalently, the  $k$ th coordinate of  $Ax$  is*

$$(Ax)_k = \sum_{j=1}^n A_{kj}x_j, \quad k = 1, \dots, m.$$

With this notation, the coordinate formula for a linear transformation is

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

**Example 12.** *Matrix–vector multiplication can be computed either as a linear combination of the columns or one coordinate at a time. For instance,*

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 14 \\ 10 \end{bmatrix},$$

*while*

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 \\ 4 \cdot 1 + 5 \cdot 2 + 6 \cdot 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \end{bmatrix}.$$

## 6.2. Finding matrix representations.

**Example 13.** Define  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x - 5y \\ 7y \end{bmatrix}.$$

With respect to the standard bases, the columns of the representing matrix are

$$T(e_1) = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad T(e_2) = \begin{bmatrix} 2 \\ -5 \\ 7 \end{bmatrix}.$$

Hence

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -5 \\ 0 & 7 \end{bmatrix}.$$

**Example 14** (Differentiation). Let  $n \geq 1$ , and let

$$T: \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}), \quad T(p) = p',$$

and use the ordered basis

$$\mathcal{B} = (1, t, t^2, \dots, t^n)$$

for both the domain and codomain. Since

$$T(1) = 0, \quad T(t) = 1, \quad T(t^j) = jt^{j-1},$$

the matrix representing  $T$  is

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & n \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

**6.3. The vector space of linear maps.** Let  $V$  and  $W$  be vector spaces over  $\mathbb{F}$ . If  $T: V \rightarrow W$  is linear and  $\alpha \in \mathbb{F}$ , define  $\alpha T: V \rightarrow W$  pointwise by

$$(\alpha T)(v) = \alpha T(v).$$

This map is again linear. Indeed, for  $u, v \in V$ ,

$$\begin{aligned} (\alpha T)(u + v) &= \alpha T(u + v) \\ &= \alpha(T(u) + T(v)) \\ &= (\alpha T)(u) + (\alpha T)(v), \end{aligned}$$

and, for  $\beta \in \mathbb{F}$ ,

$$(\alpha T)(\beta u) = \alpha T(\beta u) = \alpha \beta T(u) = \beta(\alpha T)(u).$$

Similarly, if  $T_1, T_2: V \rightarrow W$  are linear, define their sum by

$$(T_1 + T_2)(v) = T_1(v) + T_2(v).$$

The map  $T_1 + T_2$  is linear because

$$\begin{aligned} (T_1 + T_2)(u + v) &= T_1(u + v) + T_2(u + v) \\ &= (T_1 + T_2)(u) + (T_1 + T_2)(v) \end{aligned}$$

and

$$(T_1 + T_2)(\alpha u) = \alpha(T_1 + T_2)(u).$$

**Definition 13** (Space of linear maps). *The collection of all linear transformations from  $V$  to  $W$  is denoted by*

$$\mathcal{L}(V, W) = \{T: V \rightarrow W : T \text{ is linear}\}.$$

**Proposition 9.** *The set  $\mathcal{L}(V, W)$ , equipped with pointwise addition and scalar multiplication, is a vector space over  $\mathbb{F}$ .*

*Proof.* The preceding calculations show that the two operations are closed in  $\mathcal{L}(V, W)$ . All vector-space identities follow pointwise from the corresponding identities in  $W$ . The additive identity is the zero transformation  $0: V \rightarrow W$ , defined by  $0(v) = 0$ , and the additive inverse of  $T$  is the map  $-T$ , defined by  $(-T)(v) = -T(v)$ .  $\square$

**6.4. Composition of linear maps.** Let  $V$ ,  $W$ , and  $Z$  be vector spaces over  $\mathbb{F}$ . Suppose that

$$T_2: V \longrightarrow W \quad \text{and} \quad T_1: W \longrightarrow Z$$

are linear. Their *composition* is the map

$$T_1 \circ T_2: V \longrightarrow Z, \quad (T_1 \circ T_2)(v) = T_1(T_2(v)).$$

**Proposition 10.** *The composition  $T_1 \circ T_2$  is linear.*

*Proof.* For  $u, v \in V$  and  $\alpha \in \mathbb{F}$ ,

$$\begin{aligned} (T_1 \circ T_2)(u + v) &= T_1(T_2(u) + T_2(v)) \\ &= T_1(T_2(u)) + T_1(T_2(v)), \end{aligned}$$

and

$$(T_1 \circ T_2)(\alpha u) = T_1(\alpha T_2(u)) = \alpha T_1(T_2(u)).$$

□

## 7. SEPTEMBER 9, 2026

**7.1. Matrix operations and linear maps.** Let  $V$  and  $W$  be vector spaces over  $\mathbb{F}$  with ordered bases

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m).$$

Suppose that  $T, S \in \mathcal{L}(V, W)$  are represented by the matrices  $A = [A_{ij}]$  and  $B = [B_{ij}]$ , respectively. Thus

$$T(v_j) = \sum_{i=1}^m A_{ij} w_i \quad \text{and} \quad S(v_j) = \sum_{i=1}^m B_{ij} w_i.$$

For each basis vector  $v_j$ ,

$$\begin{aligned} (T + S)(v_j) &= T(v_j) + S(v_j) \\ &= \sum_{i=1}^m A_{ij} w_i + \sum_{i=1}^m B_{ij} w_i \\ &= \sum_{i=1}^m (A_{ij} + B_{ij}) w_i. \end{aligned}$$

Consequently, the matrix representing  $T + S$  is obtained by adding the matrices representing  $T$  and  $S$  entrywise.

**Definition 14** (Matrix addition). *If  $A, B \in M_{m \times n}(\mathbb{F})$ , their sum is the matrix  $A + B \in M_{m \times n}(\mathbb{F})$  defined by*

$$(A + B)_{ij} = A_{ij} + B_{ij}.$$

Similarly, if  $\alpha \in \mathbb{F}$ , then

$$(\alpha T)(v_j) = \alpha T(v_j) = \sum_{i=1}^m \alpha A_{ij} w_i.$$

Thus scalar multiplication of linear maps corresponds to entrywise scalar multiplication of matrices.

**Definition 15** (Scalar multiplication of matrices). *If  $A \in M_{m \times n}(\mathbb{F})$  and  $\alpha \in \mathbb{F}$ , then  $\alpha A \in M_{m \times n}(\mathbb{F})$  is defined by*

$$(\alpha A)_{ij} = \alpha A_{ij}.$$

**7.2. Composition and matrix multiplication.** Let  $Z$  be a third vector space with ordered basis

$$\mathcal{D} = (z_1, \dots, z_p),$$

and consider linear maps

$$T_2: V \longrightarrow W \quad \text{and} \quad T_1: W \longrightarrow Z.$$

Suppose  $T_2$  is represented by  $B = [B_{kj}] \in M_{m \times n}(\mathbb{F})$  and  $T_1$  is represented by  $A = [A_{ik}] \in M_{p \times m}(\mathbb{F})$ . For  $j = 1, \dots, n$ ,

$$\begin{aligned} (T_1 \circ T_2)(v_j) &= T_1 \left( \sum_{k=1}^m B_{kj} w_k \right) \\ &= \sum_{k=1}^m B_{kj} T_1(w_k) \\ &= \sum_{k=1}^m B_{kj} \sum_{i=1}^p A_{ik} z_i \\ &= \sum_{i=1}^p \left( \sum_{k=1}^m A_{ik} B_{kj} \right) z_i. \end{aligned}$$

The coefficient of  $z_i$  in this expression is the  $(i, j)$  entry of the matrix representing  $T_1 \circ T_2$ .

**Definition 16** (Matrix multiplication). *If  $A \in M_{p \times m}(\mathbb{F})$  and  $B \in M_{m \times n}(\mathbb{F})$ , their product is the matrix  $AB \in M_{p \times n}(\mathbb{F})$  whose entries are*

$$(AB)_{ij} = \sum_{k=1}^m A_{ik}B_{kj}.$$

Thus composition of linear maps corresponds to multiplication of their representing matrices, in the same order:

$$[T_1 \circ T_2] = [T_1][T_2].$$

Whenever all the products and sums involved are defined, matrix multiplication satisfies

$$\begin{aligned} (AB)C &= A(BC), \\ A(B + C) &= AB + AC, \\ (A + B)C &= AC + BC, \\ A(\alpha B) &= \alpha(AB) = (\alpha A)B. \end{aligned}$$

In general, however,  $AB \neq BA$ ; the two products may even have different sizes, or one may be undefined. In particular,  $M_{n \times n}(\mathbb{F})$ , with matrix addition, scalar multiplication, and matrix multiplication, is an associative algebra over  $\mathbb{F}$ .

### 7.3. Vector subspaces.

**Definition 17** (Subspace). *Let  $V$  be a vector space over  $\mathbb{F}$ . A subset  $U \subseteq V$  is a subspace of  $V$  if  $U$ , with the addition and scalar multiplication inherited from  $V$ , is itself a vector space over  $\mathbb{F}$ .*

Checking all the vector-space axioms is unnecessary. The following criterion reduces the verification to three conditions.

**Proposition 11** (Subspace criterion). *A subset  $U \subseteq V$  is a subspace of  $V$  if and only if*

- (i)  $0 \in U$ ;
- (ii)  $u + v \in U$  whenever  $u, v \in U$ ;
- (iii)  $\alpha u \in U$  whenever  $u \in U$  and  $\alpha \in \mathbb{F}$ .

Equivalently, if  $U$  is known to be nonempty, it is enough to check the two closure conditions (ii) and (iii).

**Example 15.** The affine line

$$H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : x = 2 \right\}$$

is not a subspace of  $\mathbb{R}^2$ , because it does not contain the zero vector.

The subspaces of  $\mathbb{R}^2$  are precisely the zero subspace  $\{0\}$ , the lines through the origin, and  $\mathbb{R}^2$  itself.

8. SEPTEMBER 11, 2026

### 8.1. The subspace criterion.

*Proof of Proposition 11.* If  $U$  is a subspace of  $V$ , then  $U$  is a vector space under the operations inherited from  $V$ . It therefore contains the additive identity and is closed under addition and scalar multiplication.

Conversely, suppose conditions (i)–(iii) hold. The additive identity belongs to  $U$  by (i). If  $u \in U$ , then condition (iii), applied with the scalar  $-1$ , gives

$$-u = (-1)u \in U.$$

Hence  $U$  contains the additive inverse of each of its elements. The remaining vector-space axioms are identities already valid in  $V$ , so they remain valid when the vectors are restricted to  $U$ . Thus  $U$  is a subspace of  $V$ .

Finally, if  $U$  is nonempty and satisfies the two closure conditions, choose  $u \in U$ . Closure under scalar multiplication gives  $0 = 0u \in U$ , so condition (i) follows automatically.  $\square$

### 8.2. Kernels and images.

**Definition 18** (Kernel and image). Let  $T \in \mathcal{L}(V, W)$ . The kernel and image of  $T$  are

$$\ker T = \{v \in V : T(v) = 0\}$$

and

$$\operatorname{im} T = \{T(v) : v \in V\},$$

respectively. The image is also called the range of  $T$ .

**Proposition 12.** *If  $T \in \mathcal{L}(V, W)$ , then  $\ker T$  is a subspace of  $V$  and  $\operatorname{im} T$  is a subspace of  $W$ .*

*Proof.* Since  $T(0) = 0$ , both sets contain the appropriate zero vector. If  $u, v \in \ker T$  and  $\alpha \in \mathbb{F}$ , then

$$T(u + v) = T(u) + T(v) = 0 \quad \text{and} \quad T(\alpha u) = \alpha T(u) = 0.$$

Thus  $\ker T$  is a subspace. If  $y_1 = T(v_1)$  and  $y_2 = T(v_2)$  are in  $\operatorname{im} T$ , then

$$y_1 + y_2 = T(v_1 + v_2) \quad \text{and} \quad \alpha y_1 = T(\alpha v_1),$$

so the image is also a subspace.  $\square$

**Example 16.** *The continuous real-valued functions on  $[0, 1]$  form a subspace*

$$C([0, 1]) \subseteq \{f : [0, 1] \rightarrow \mathbb{R}\}$$

*of the vector space of all real-valued functions on  $[0, 1]$ .*

**Example 17.** *Fix  $b \in \mathbb{F}$  and let*

$$U_b = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{F}^4 : x_3 = 5x_4 + b \right\}.$$

*Then  $U_b$  is a subspace of  $\mathbb{F}^4$  if and only if  $b = 0$ . Indeed, the zero vector belongs to  $U_b$  only when  $b = 0$ , and when  $b = 0$  the defining equation  $x_3 = 5x_4$  is preserved by addition and scalar multiplication.*

### 8.3. Sums of subspaces.

**Definition 19** (Sum of subspaces). *If  $V_1, \dots, V_m$  are subspaces of  $V$ , their sum is*

$$V_1 + \cdots + V_m = \{v_1 + \cdots + v_m : v_i \in V_i \text{ for } i = 1, \dots, m\}.$$

**Example 18.** *Let*

$$U = \left\{ \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} : x \in \mathbb{F} \right\}, \quad W = \left\{ \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix} : y \in \mathbb{F} \right\}.$$

*Then*

$$U + W = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{F} \right\}.$$

**Proposition 13.** *If  $V_1, \dots, V_m$  are subspaces of  $V$ , then  $V_1 + \dots + V_m$  is the smallest subspace of  $V$  containing each  $V_i$ .*

*Proof.* The sum contains  $0 = 0 + \dots + 0$ . Adding two elements of the sum, or multiplying one by a scalar, can be done componentwise inside each  $V_i$ , so the sum is a subspace. For each  $i$ , any  $v_i \in V_i$  can be written as

$$v_i = 0 + \dots + 0 + v_i + 0 + \dots + 0,$$

and hence  $V_i \subseteq V_1 + \dots + V_m$ .

Conversely, let  $W$  be any subspace of  $V$  containing all the  $V_i$ . If  $v = v_1 + \dots + v_m$  with  $v_i \in V_i$ , then every  $v_i$  belongs to  $W$ . Since  $W$  is closed under addition,  $v \in W$ . Therefore  $V_1 + \dots + V_m \subseteq W$ .  $\square$

#### 8.4. Direct sums.

**Definition 20** (Direct sum). *Let  $V_1, \dots, V_m$  be subspaces of  $V$ . Their sum is a direct sum if every element of  $V_1 + \dots + V_m$  has a unique representation*

$$v = v_1 + \dots + v_m, \quad v_i \in V_i.$$

*In this case we write*

$$V_1 \oplus \dots \oplus V_m.$$

**Proposition 14** (Zero-sum criterion). *The sum  $V_1 + \dots + V_m$  is direct if and only if the equation*

$$0 = v_1 + \dots + v_m, \quad v_i \in V_i,$$

*implies  $v_1 = \dots = v_m = 0$ .*

*Proof.* If the sum is direct, then  $0 = 0 + \cdots + 0$  is one representation of the zero vector, so uniqueness forces every other representation to have all components equal to zero.

Conversely, suppose the zero vector has only the trivial representation. If

$$v_1 + \cdots + v_m = u_1 + \cdots + u_m,$$

where  $v_i, u_i \in V_i$ , then

$$0 = (v_1 - u_1) + \cdots + (v_m - u_m).$$

The hypothesis gives  $v_i - u_i = 0$  for each  $i$ , so  $v_i = u_i$ . Thus every representation is unique.  $\square$

**Proposition 15.** *If  $U$  and  $W$  are subspaces of  $V$ , then*

$$U + W = U \oplus W \iff U \cap W = \{0\}.$$

*Proof.* Suppose first that  $U + W$  is a direct sum, and let  $v \in U \cap W$ . Then

$$0 = v + (-v) = 0 + 0$$

gives two representations with one term in  $U$  and one term in  $W$ . Uniqueness implies  $v = 0$ .

Conversely, assume  $U \cap W = \{0\}$ . If

$$u + w = u' + w', \quad u, u' \in U, \quad w, w' \in W,$$

then

$$u - u' = w' - w \in U \cap W.$$

Hence  $u = u'$  and  $w = w'$ , so the representation is unique.  $\square$

For the coordinate-axis subspaces in the preceding example,  $U \cap W = \{0\}$ , and hence their sum is  $U \oplus W$ .

### 8.5. Product vector spaces.

**Definition 21** (Product vector space). *Let  $V_1, \dots, V_m$  be vector spaces over  $\mathbb{F}$ . Their Cartesian product is*

$$V_1 \times \cdots \times V_m = \{(v_1, \dots, v_m) : v_i \in V_i\}.$$

Addition and scalar multiplication are defined coordinatewise:

$$\begin{aligned}(v_1, \dots, v_m) + (u_1, \dots, u_m) &= (v_1 + u_1, \dots, v_m + u_m), \\ \alpha(v_1, \dots, v_m) &= (\alpha v_1, \dots, \alpha v_m).\end{aligned}$$

With these operations,  $V_1 \times \dots \times V_m$  is a vector space over  $\mathbb{F}$ ; all vector-space axioms hold coordinatewise.

**Example 19.** In  $\mathcal{P}_5(\mathbb{R}) \times \mathbb{R}^3$ ,

$$\begin{aligned}\left(5 - 6t + 4t^2, \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix}\right) + \left(t + 9t^5, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}\right) \\ = \left(5 - 5t + 4t^2 + 9t^5, \begin{bmatrix} 5 \\ 10 \\ 9 \end{bmatrix}\right).\end{aligned}$$

9. SEPTEMBER 14, 2026

**9.1. Direct sums and injectivity.** Let  $V_1, \dots, V_m$  be subspaces of a vector space  $V$ . Define the summation map

$$\Sigma: V_1 \times \dots \times V_m \longrightarrow V_1 + \dots + V_m$$

by

$$\Sigma(v_1, \dots, v_m) = v_1 + \dots + v_m.$$

This map is linear and surjective.

**Proposition 16.** *The sum  $V_1 + \dots + V_m$  is direct if and only if the summation map  $\Sigma$  is injective.*

*Proof.* The map  $\Sigma$  is injective precisely when

$$\Sigma(v_1, \dots, v_m) = \Sigma(u_1, \dots, u_m)$$

implies

$$(v_1, \dots, v_m) = (u_1, \dots, u_m).$$

Equivalently, every vector in  $V_1 + \dots + V_m$  has only one representation as a sum of vectors from the respective subspaces. This is exactly the definition of a direct sum.  $\square$

**9.2. Quotient vector spaces.** Let  $U$  be a subset of  $V$ . For  $v \in V$ , define

$$v + U = \{v + u : u \in U\}.$$

The set  $v + U$  is called a *translate* of  $U$ . When  $U$  is a subspace, a translate is also called a *coset* of  $U$ .

**Example 20.** *Let*

$$U = \left\{ \begin{bmatrix} x \\ 2x \end{bmatrix} : x \in \mathbb{R} \right\} \subseteq \mathbb{R}^2.$$

*Then*

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + U = \left\{ \begin{bmatrix} 2+x \\ 1+2x \end{bmatrix} : x \in \mathbb{R} \right\}.$$

*The original subspace is the line  $y = 2x$ , while this translate is the parallel line  $y = 2x - 3$ .*

**Definition 22** (Quotient space). *If  $U$  is a subspace of  $V$ , the quotient space of  $V$  by  $U$  is the set of all cosets of  $U$ :*

$$V/U = \{v + U : v \in V\}.$$

For the subspace  $U \subseteq \mathbb{R}^2$  in the preceding example,  $\mathbb{R}^2/U$  is the collection of all lines in  $\mathbb{R}^2$  with slope 2.

**Proposition 17.** *Let  $U$  be a subspace of  $V$  and let  $v, w \in V$ . The following are equivalent:*

- (i)  $v - w \in U$ ;
- (ii)  $v + U = w + U$ ;
- (iii)  $(v + U) \cap (w + U) \neq \emptyset$ .

*Proof.* Suppose first that  $v - w \in U$ . If  $v + u \in v + U$ , then

$$v + u = w + (v - w + u) \in w + U,$$

because  $v - w + u \in U$ . Thus  $v + U \subseteq w + U$ . Since  $w - v = -(v - w) \in U$ , the reverse inclusion follows in the same way, so  $v + U = w + U$ .

If  $v + U = w + U$ , then the two cosets plainly have nonempty intersection.

Finally, suppose  $(v + U) \cap (w + U) \neq \emptyset$ . There are  $u_1, u_2 \in U$  such that

$$v + u_1 = w + u_2.$$

Therefore

$$v - w = u_2 - u_1 \in U.$$

□

Thus two cosets of a subspace are either equal or disjoint. We define addition and scalar multiplication on  $V/U$  by

$$\begin{aligned}(v + U) + (w + U) &= (v + w) + U, \\ \alpha(v + U) &= \alpha v + U.\end{aligned}$$

Proposition 17 shows that these operations do not depend on the choice of representatives. With these operations,  $V/U$  is a vector space over  $\mathbb{F}$ , whose zero vector is the coset  $U = 0 + U$ .

### 9.3. Finite-dimensional vector spaces.

**Definition 23** (Finite-dimensional vector space). *A vector space  $V$  is finite-dimensional if some finite collection of vectors spans  $V$ . It is infinite-dimensional if it is not finite-dimensional.*

**Example 21.** *The coordinate space  $\mathbb{F}^n$  is finite-dimensional because*

$$\mathbb{F}^n = \text{span}\{e_1, \dots, e_n\}.$$

*Likewise, the space of polynomials of degree at most  $m$  is finite-dimensional because*

$$\mathcal{P}_m(\mathbb{F}) = \text{span}\{1, t, t^2, \dots, t^m\}.$$

**Example 22.** *The vector space  $\mathcal{P}(\mathbb{F})$  of all polynomials with coefficients in  $\mathbb{F}$  is infinite-dimensional. Indeed, suppose a finite collection of polynomials spanned  $\mathcal{P}(\mathbb{F})$ , and let  $m$  be the largest degree among the nonzero polynomials in the collection. Every polynomial in their span would have degree at most  $m$ , so  $t^{m+1}$  could not belong to the span, a contradiction.*

10. SEPTEMBER 16, 2026

**10.1. Comparing independent and spanning collections.**

**Proposition 18.** *Suppose that  $u_1, \dots, u_m$  is a linearly independent collection in  $V$  and that  $w_1, \dots, w_n$  spans  $V$ . Then*

$$m \leq n.$$

*Thus, in a finite-dimensional vector space, every linearly independent collection has cardinality at most that of every finite spanning collection.*

*Proof.* We successively insert the vectors  $u_1, u_2, \dots$  into the spanning list  $w_1, \dots, w_n$ , removing one of the  $w$ 's after each insertion.

First adjoin  $u_1$  to obtain

$$u_1, w_1, \dots, w_n.$$

This collection is linearly dependent because  $w_1, \dots, w_n$  spans  $V$ . Since  $u_1 \neq 0$ , one of the  $w$ 's is a linear combination of the other vectors. Remove that vector. The resulting collection still spans  $V$  and has  $n$  vectors.

Inductively, suppose that after  $k-1$  steps we have a spanning collection of  $n$  vectors consisting of  $u_1, \dots, u_{k-1}$  and  $n-k+1$  of the original  $w$ 's. Adjoin  $u_k$ . The resulting collection is dependent. In a nontrivial dependence relation, some remaining  $w$  must have a nonzero coefficient; otherwise the relation would contradict the linear independence of  $u_1, \dots, u_k$ . We can therefore remove that  $w$  without changing the span.

If  $m > n$ , then after  $n$  steps the collection  $u_1, \dots, u_n$  would span  $V$ . In particular,  $u_{n+1}$  would lie in their span, contradicting the linear independence of  $u_1, \dots, u_m$ . Hence  $m \leq n$ .  $\square$

**Example 23.** *No collection of four vectors in  $\mathbb{R}^3$  is linearly independent, because the three standard basis vectors span  $\mathbb{R}^3$ . Similarly, no collection of three vectors can span  $\mathbb{R}^4$ , because the four standard basis vectors are linearly independent in  $\mathbb{R}^4$ .*

## 10.2. Subspaces of finite-dimensional spaces.

**Proposition 19.** *Every subspace of a finite-dimensional vector space is finite-dimensional.*

*Proof.* Let  $U$  be a subspace of a finite-dimensional vector space  $V$ . If  $U = \{0\}$ , then the empty collection spans  $U$ , so  $U$  is finite-dimensional.

Otherwise choose a nonzero vector  $u_1 \in U$ . If  $U = \text{span}\{u_1\}$ , we are done. If not, choose

$$u_2 \in U \setminus \text{span}\{u_1\}.$$

Continue in this fashion: once  $u_1, \dots, u_{k-1}$  have been chosen, stop if they span  $U$ ; otherwise choose

$$u_k \in U \setminus \text{span}\{u_1, \dots, u_{k-1}\}.$$

At every stage the chosen collection is linearly independent. Since  $V$  has a finite spanning collection, Proposition 18 bounds the number of vectors in any linearly independent collection in  $V$ . The process must therefore stop after finitely many steps, at which point the chosen vectors span  $U$ .  $\square$

**Corollary 1.** *Every finite-dimensional vector space has a basis.*

*Proof.* By definition, a finite-dimensional vector space has a finite spanning collection. Removing redundant vectors from that collection produces a basis.  $\square$

## 10.3. Extending a linearly independent collection.

**Proposition 20** (Basis extension theorem). *Every finite linearly independent collection in a finite-dimensional vector space can be extended to a basis of the space.*

*Proof.* Let  $u_1, \dots, u_m$  be linearly independent in  $V$ , and choose a finite spanning collection  $w_1, \dots, w_n$  for  $V$ . The combined collection

$$u_1, \dots, u_m, w_1, \dots, w_n$$

spans  $V$ . Examine the  $w$ 's in order. Whenever a vector  $w_j$  lies in the span of the vectors preceding it, remove it. Removing such a vector

does not change the span, and none of the  $u_i$  is removed. After all the  $w$ 's have been considered, the remaining collection spans  $V$  and is linearly independent. It is therefore a basis containing  $u_1, \dots, u_m$ .  $\square$

**Example 24.** Consider the linearly independent vectors

$$v_1 = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -9 \\ 6 \\ 8 \end{bmatrix}$$

in  $\mathbb{R}^3$ . Adjoin the standard basis vectors  $e_1, e_2, e_3$ . Since

$$e_1 = -\frac{1}{5}(v_2 - 2v_1),$$

the vector  $e_1$  is redundant. On the other hand,  $e_2$  is not in  $\text{span}\{v_1, v_2\}$ , so we retain it. Finally,

$$e_3 = -\frac{1}{10} \left( v_2 - \frac{9}{2}v_1 + \frac{15}{2}e_2 \right),$$

so  $e_3$  is redundant. Therefore

$$\left\{ \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -9 \\ 6 \\ 8 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

is a basis of  $\mathbb{R}^3$  extending the original collection.

11. SEPTEMBER 18, 2026

### 11.1. Complements of subspaces.

**Proposition 21.** Suppose that  $V$  is finite-dimensional and that  $U$  is a subspace of  $V$ . Then there is a subspace  $W$  of  $V$  such that

$$V = U \oplus W.$$

*Proof.* Because  $V$  is finite-dimensional, so is  $U$ . Let  $u_1, \dots, u_m$  be a basis of  $U$ . This collection is linearly independent in  $V$ , so the basis extension theorem gives a basis

$$u_1, \dots, u_m, w_1, \dots, w_n$$

of  $V$ . Let

$$W = \text{span}\{w_1, \dots, w_n\}.$$

Certainly  $U + W \subseteq V$ . Conversely, if  $v \in V$ , then there are scalars  $a_1, \dots, a_m, b_1, \dots, b_n$  such that

$$v = \underbrace{a_1u_1 + \dots + a_mu_m}_{\in U} + \underbrace{b_1w_1 + \dots + b_nw_n}_{\in W}.$$

Hence  $v \in U + W$ , and therefore  $V = U + W$ .

It remains to show that  $U \cap W = \{0\}$ . Let  $v \in U \cap W$ . Then

$$v = a_1u_1 + \dots + a_mu_m = b_1w_1 + \dots + b_nw_n$$

for some scalars  $a_i$  and  $b_j$ . Thus

$$a_1u_1 + \dots + a_mu_m - b_1w_1 - \dots - b_nw_n = 0.$$

Because the combined collection is a basis of  $V$ , all the coefficients are zero. In particular,  $v = 0$ , so  $U \cap W = \{0\}$ . Thus  $V = U \oplus W$ .  $\square$

## 11.2. Dimension.

**Proposition 22.** *Any two bases of a finite-dimensional vector space have the same cardinality.*

*Proof.* Let  $\mathcal{B}_1$  and  $\mathcal{B}_2$  be bases of a finite-dimensional vector space  $V$ . Because  $\mathcal{B}_1$  is linearly independent and  $\mathcal{B}_2$  spans  $V$ , Proposition 18 gives

$$|\mathcal{B}_1| \leq |\mathcal{B}_2|.$$

Interchanging the roles of the two bases gives

$$|\mathcal{B}_2| \leq |\mathcal{B}_1|.$$

Therefore  $|\mathcal{B}_1| = |\mathcal{B}_2|$ .  $\square$

**Definition 24** (Dimension). *The dimension of a finite-dimensional vector space  $V$  is the cardinality of any basis of  $V$ . It is denoted by  $\dim V$ .*

**Example 25.** The standard basis  $\{e_1, \dots, e_n\}$  of  $\mathbb{F}^n$  has  $n$  elements, so

$$\dim \mathbb{F}^n = n.$$

**Example 26.** The standard basis  $\{1, x, \dots, x^m\}$  of  $\mathcal{P}_m(\mathbb{F})$  has  $m + 1$  elements, so

$$\dim \mathcal{P}_m(\mathbb{F}) = m + 1.$$

**Example 27.** Let

$$U = \left\{ \begin{bmatrix} x \\ 0 \\ y \end{bmatrix} : x, y \in \mathbb{F} \right\}.$$

Because

$$U = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

and the two displayed vectors are linearly independent, they form a basis of  $U$ . Hence  $\dim U = 2$ .

**Example 28.** As a vector space over  $\mathbb{R}$ , the space  $\mathbb{R}^2$  has dimension 2. As a vector space over  $\mathbb{C}$ , the space  $\mathbb{C}$  has dimension 1. Although  $\mathbb{R}^2$  and  $\mathbb{C}$  are often identified as sets, the choice of scalar field is essential when determining dimension.

### 11.3. Consequences of dimension.

**Proposition 23.** If  $V$  is finite-dimensional and  $U$  is a subspace of  $V$ , then

$$\dim U \leq \dim V.$$

*Proof.* A basis of  $U$  is linearly independent in  $V$ , while a basis of  $V$  spans  $V$ . The result follows from Proposition 18.  $\square$

**Proposition 24.** Suppose that  $V$  is finite-dimensional. Every linearly independent collection of  $\dim V$  vectors in  $V$  is a basis of  $V$ .

*Proof.* Write  $\dim V = n$ , and suppose that  $v_1, \dots, v_n$  is linearly independent in  $V$ . By the basis extension theorem, this collection can be

extended to a basis of  $V$ . Every basis of  $V$  has exactly  $n$  vectors, however, so no vectors need to be adjoined. Thus  $v_1, \dots, v_n$  is already a basis of  $V$ .  $\square$

**Corollary 2.** *Suppose that  $V$  is finite-dimensional and that  $U$  is a subspace of  $V$ . If  $\dim U = \dim V$ , then  $U = V$ .*

*Proof.* Let  $u_1, \dots, u_n$  be a basis of  $U$ . Then  $n = \dim U = \dim V$ , and  $u_1, \dots, u_n$  is linearly independent in  $V$ . By Proposition 24, it is a basis of  $V$ . Hence every vector in  $V$  is a linear combination of vectors in  $U$ , so  $V \subseteq U$ . The reverse inclusion holds because  $U$  is a subspace of  $V$ . Therefore  $U = V$ .  $\square$

**Example 29.** *Let*

$$U = \{p \in \mathcal{P}_3(\mathbb{R}) : p'(5) = 0\}.$$

*We find a basis of  $U$ . Write*

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

*Because*

$$p'(x) = a_1 + 2a_2x + 3a_3x^2,$$

*the condition  $p'(5) = 0$  is equivalent to*

$$a_1 + 10a_2 + 75a_3 = 0, \quad \text{or equivalently} \quad a_1 = -10a_2 - 75a_3.$$

*Therefore every  $p \in U$  has the form*

$$\begin{aligned} p(x) &= a_0 + (-10a_2 - 75a_3)x + a_2x^2 + a_3x^3 \\ &= a_0 + a_2(x^2 - 10x) + a_3(x^3 - 75x). \end{aligned}$$

*Thus the collection*

$$\{1, x^2 - 10x, x^3 - 75x\}$$

*spans  $U$ . It is linearly independent because its members have distinct degrees. Consequently,*

$$\boxed{\{1, x^2 - 10x, x^3 - 75x\}}$$

*is a basis of  $U$ , and  $\dim U = 3$ .*

12. SEPTEMBER 21, 2026

**12.1. Spanning collections and dimension.**

**Proposition 25.** *Suppose that  $V$  is finite-dimensional. Every collection of  $\dim V$  vectors that spans  $V$  is a basis of  $V$ .*

*Proof.* Write  $\dim V = n$ , and suppose that  $v_1, \dots, v_n$  spans  $V$ . This collection can be reduced to a basis by removing redundant vectors. However, every basis of  $V$  has exactly  $n$  vectors, so no vectors can be removed. Thus  $v_1, \dots, v_n$  is already a basis of  $V$ .  $\square$

**12.2. Dimension of a sum of subspaces.**

**Proposition 26.** *If  $V_1$  and  $V_2$  are subspaces of a finite-dimensional vector space, then*

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

*Proof.* Let  $v_1, \dots, v_m$  be a basis of  $V_1 \cap V_2$ , so that  $\dim(V_1 \cap V_2) = m$ . This collection is linearly independent in  $V_1$ , so the basis extension theorem gives a basis

$$v_1, \dots, v_m, u_1, \dots, u_j$$

of  $V_1$ . Similarly, extend  $v_1, \dots, v_m$  to a basis

$$v_1, \dots, v_m, w_1, \dots, w_k$$

of  $V_2$ . Hence

$$\dim V_1 = m + j \quad \text{and} \quad \dim V_2 = m + k.$$

We will show that

$$v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$$

is a basis of  $V_1 + V_2$ .

All the displayed vectors belong to  $V_1 + V_2$ , so their span is contained in  $V_1 + V_2$ . Conversely, every vector in  $V_1 + V_2$  is a sum of a vector in  $V_1$  and a vector in  $V_2$ . Expressing each summand in the corresponding basis shows that the combined collection spans  $V_1 + V_2$ .

To prove linear independence, suppose that

$$(1) \quad a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j + c_1w_1 + \cdots + c_kw_k = 0.$$

Rearranging gives

$$c_1w_1 + \cdots + c_kw_k = -a_1v_1 - \cdots - a_mv_m - b_1u_1 - \cdots - b_ju_j.$$

The right-hand side belongs to  $V_1$ , while the left-hand side belongs to  $V_2$ . Therefore

$$c_1w_1 + \cdots + c_kw_k \in V_1 \cap V_2.$$

Since  $v_1, \dots, v_m$  is a basis of the intersection, there are scalars  $d_1, \dots, d_m$  such that

$$c_1w_1 + \cdots + c_kw_k = d_1v_1 + \cdots + d_mv_m.$$

Thus

$$c_1w_1 + \cdots + c_kw_k - d_1v_1 - \cdots - d_mv_m = 0.$$

The collection  $v_1, \dots, v_m, w_1, \dots, w_k$  is a basis of  $V_2$  and hence linearly independent. Consequently, all the  $c$ 's and  $d$ 's are zero. Equation (1) now becomes

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0.$$

Because  $v_1, \dots, v_m, u_1, \dots, u_j$  is a basis of  $V_1$ , all the  $a$ 's and  $b$ 's are also zero. This proves linear independence.

The combined collection is therefore a basis of  $V_1 + V_2$ , and

$$\begin{aligned} \dim(V_1 + V_2) &= m + j + k \\ &= (m + j) + (m + k) - m \\ &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2). \end{aligned}$$

□

### 12.3. Linear maps on finite-dimensional spaces.

**Theorem 2** (Fundamental theorem of linear maps). *Suppose that  $V$  is finite-dimensional and that  $T \in \mathcal{L}(V, W)$ . Then  $\text{im } T$  is finite-dimensional and*

$$\dim V = \dim \ker T + \dim \text{im } T.$$

Recall that

$$\ker T = \{v \in V : T(v) = 0\} \quad \text{and} \quad \operatorname{im} T = \{T(v) : v \in V\}.$$

Before proving the theorem, we connect the kernel and image with injectivity and surjectivity, beginning with injectivity.

**Proposition 27.** *Let  $T \in \mathcal{L}(V, W)$ . Then  $T$  is injective if and only if  $\ker T = \{0\}$ .*

*Proof.* First note that every linear map sends zero to zero, since

$$T(0) = T(0 + 0) = T(0) + T(0)$$

implies  $T(0) = 0$ . This does not require injectivity. In particular,  $\{0\} \subseteq \ker T$ .

Suppose that  $T$  is injective. If  $v \in \ker T$ , then

$$T(v) = 0 = T(0).$$

Injectivity gives  $v = 0$ , so  $\ker T \subseteq \{0\}$ . Therefore  $\ker T = \{0\}$ .

Conversely, suppose that  $\ker T = \{0\}$ . Let  $v_1, v_2 \in V$  satisfy  $T(v_1) = T(v_2)$ . By linearity,

$$T(v_1 - v_2) = T(v_1) - T(v_2) = 0.$$

Hence  $v_1 - v_2 \in \ker T = \{0\}$ , so  $v_1 = v_2$ . Thus  $T$  is injective.  $\square$